

Study of a vaporizing water micro-thruster

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Abstract

Vaporizing water micro-thrusters are fabricated and tested. A single micro-thruster fabricated using MEMS technology consists of a micro-resistor, a vaporizing chamber, a nozzle, a propellant inlet and a micro channel. The water propellant is fed into the thruster from a propellant tank by capillary force and pressure. The micro-thruster is designed to operate in pulse mode. During each period, an electric pulse is applied to the micro-resistor to heat the water in the chamber to vaporize it into high-pressure gas. A thrust is then produced as the gas exits through the nozzle. Test results show that for a single micro-thruster with pulse power of 30 W, the total impulse produced in a second is more than 0.2×10^{-6} N s. © 2001 Elsevier Science B.V. All rights reserved.

Keywords: Thruster; Micro-machining; Resistojet; Micro-propulsion

1. Introduction

A micro-propulsion system can be used in micro satellites for orbit lifting, speed adjusting, gravitation compensation, station keeping, attitude control and so on. It mainly consists of a nozzle, a micro combustion chamber, micro valve(s), micro pump(s), micro channel(s), micro pressure and temperature sensors and a control circuit, etc. The current research on micro-propulsion systems is to develop a small chemical system with high performance and a low power-consuming electric system. JPL exhibited prototypes of two kinds of micro-resistojet fabricated using MEMS technology in 1997 [1]. One is a subliming solid micro-thruster, the other a vaporizing liquid micro-thruster. Each of them weights only a few grams with a size of 1 cm^2 . Their performance figures include an Isp up to 50–125 s, a thrust up to 0.5 mN, and a power less than 5 W. TRW, the aerospace corporation and CIT have developed and tested “Digital Propulsion” arrays [2]. A three-layer sandwich structure is fabricated, containing micro-resistors, thrust chambers, and rupture diaphragms. An impulsion of 10^{-4} N s and a power of about 100 W of power have reached in initial tests. A work is currently underway to study a micro-bipropellant rocket engine in MIT. The engine is composed of five or six silicon wafers piled up and interconnected using microfabrication technology. A combustion chamber, a nozzle, two micro-pumps, two micro-valves and

a cooling channel will be fabricated on silicon wafers. It is expected to reach a 15 N of thrust while consuming liquid oxygen and methanol propellants [3].

In this paper, a vaporizing water micro-thruster with a heating resistor is proposed. The fabrication process and the characteristics testing of the micro-thruster are described.

2. Operation principle

The micro-thruster we developed is mainly composed of a heating resistor, a vaporizing chamber, a nozzle, a propellant inlet and a micro channel. The propellant is fed into the chamber by capillary effect and pressure from a propellant storing tank through the propellant inlet and the micro channel. The micro-thruster is designed to operate in pulse mode. A pulsed electric current, with high voltage but very short pulse duration of a few microseconds, is applied to the heating resistor. Consequently, the propellant in the chamber is vaporized into gas with high temperature and high pressure in a very short time. Then the gas exits through the specially shaped nozzle with high speed, and thus producing thrust in a reversed direction.

As shown in Fig. 1, the micro resistojet propulsion system [4,5] consists of a power source, a control circuit, a propellant control system and a single or arrayed vaporizing water micro-thruster(s).

3. Configuration and fabrication

The micro-thruster chip, fabricated with MEMS technology, consists of two silicon wafers as shown in Fig. 2. The

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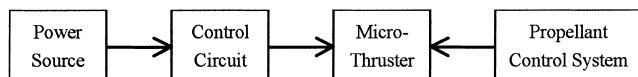


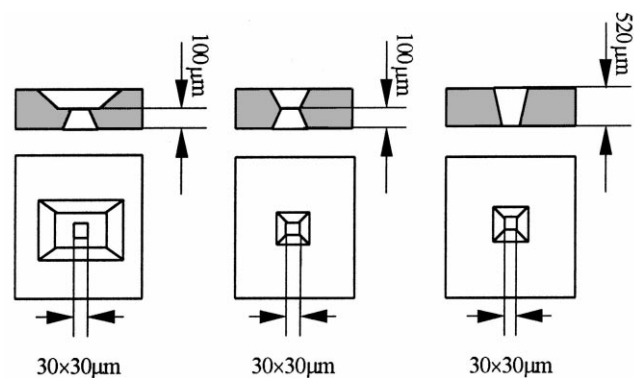
Fig. 1. Construction of a micro resistojet propulsion system.

nozzle and the holes for wire bonding are fabricated on the top wafer by bulk silicon etching. The vaporizing chamber and the micro channel with a depth of $2\text{ }\mu\text{m}$ is etched into bulk silicon on the front side of the bottom wafer. The heating resistor, internal leads and bonding pads are formed by metal Ti with 200 nm thickness at the bottom of the chamber to simplify the fabrication process. The propellant inlet and a cavity are etched into the bulk silicon from the back side of the bottom wafer. The cavity formed under the heating resistor is aimed to reduce heat capacitance of the bottom silicon layer, which in turn decreases the heat loss. The vaporizing chamber is connected with a propellant tank through a micro channel and an inlet. Wires are bonded with the pads through the holes in the top wafer, and the holes are filled with glue to seal water. Three types of nozzles are designed: convergent nozzle, de laval nozzle, and divergent nozzle, which are shown in Fig. 3.

The fabrication process of a micro-thruster chip is shown in Fig. 4, in which (a) is the process of the top wafer and (b) the bottom wafer. After the process and wafer dicing, a top and a bottom chip are bonded together by glue to form a micro-thruster chip.

In some micro-thrusters, the inside surface of the nozzle is specially treated to be hydrophobic to prevent water propellant from entering the nozzle to reduce the water drop ejection in operation, which otherwise wastes the propellant and degrades the performance of the micro-thruster significantly. A hydrophobic film is formed on the wafer by spin coating hot poly (vinylidene fluoride) (PVDF) solution at 230°C and heating the wafer. PVDF's surface tension is 0.028 N/m [6].

We fabricated individual micro-thrusters and 2×2 arrayed micro-thrusters of different structures and sizes.



(a) convergent nozzle (b) de laval nozzle (c) divergent nozzle

Fig. 3. Schematic diagrams of three types of nozzles: (a) convergent nozzle; (b) de laval nozzle; (c) divergent nozzle.

The arrayed one is able to work with only one thruster or with several thrusters at the same time. Fig. 5(a) shows an SEM graph of the top silicon wafer of a single micro-thruster. There is a nozzle with a throat size of $30\text{ }\mu\text{m} \times 30\text{ }\mu\text{m}$ on the bottom of the center rectangular slot. The two square holes are for wire bonding. Fig. 5(b) shows the front side of the bottom silicon wafer. The shape of the metal Ti layer consisting of heating resistor, the wide internal leads and the bonding pads, on the bottom of the chamber, is discernible. Two sizes of the heating resistors were fabricated, which are shown in Fig. 6(a) and (b), respectively. The inlet with the throat size of $120\text{ }\mu\text{m} \times 120\text{ }\mu\text{m}$ and the micro channel with $900\text{ }\mu\text{m}$ in length and $200\text{ }\mu\text{m}$ in width for propellant input are also shown. Fig. 5(c) shows the back side of the bottom silicon wafer. The rectangular cavity with $420\text{ }\mu\text{m}$ in depth is used to reduce heat capacitance, and the upper square hole is the inlet. Fig. 7 is a graph of a packaged 2×2 micro-thruster array.

Units (80) are fabricated on a pair of 4 in. silicon wafers. The size of the silicon chip of the individual micro-thruster is $7\text{ mm} \times 7\text{ mm} \times 1\text{ mm}$, and the packaged chip is $18\text{ mm} \times$

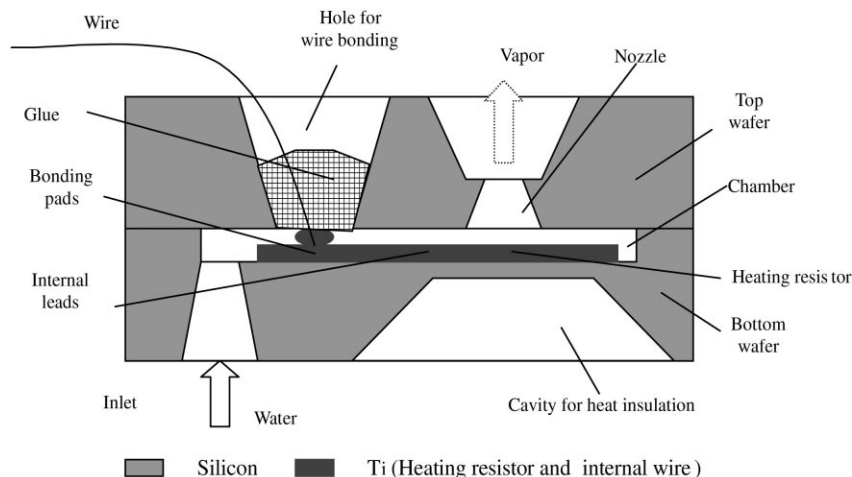
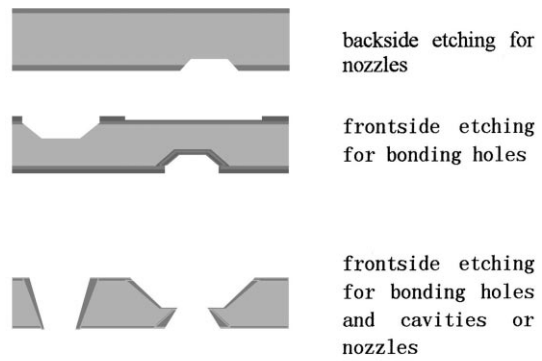
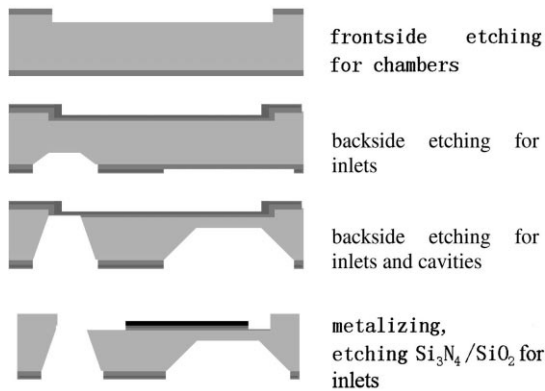


Fig. 2. Schematic configuration of the micro-thruster chip.



Si SiO₂ Si₃N₄

(a) Fabrication process of the top wafer



Si SiO₂ Si₃N₄ Ti

(b) Fabrication process of the bottom wafer

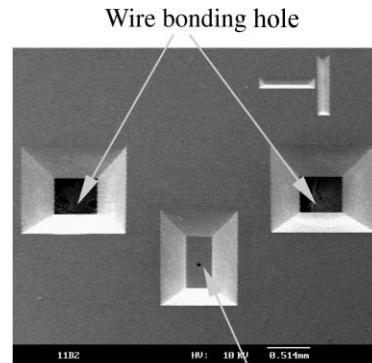
Fig. 4. Fabrication process of the micro-thruster chip: (a) fabrication process of the top wafer; (b) fabrication process of the bottom wafer.

13 mm × 5 mm. The size of the silicon chip of the 2 × 2 arrayed micro-thrusters is 10 mm × 10 mm × 1 mm, and the packaged chip is 28 mm × 20 mm × 5 mm. The weight of the packaged chips of the both micro-thrusters ranges from 0.9 to 2.1 g.

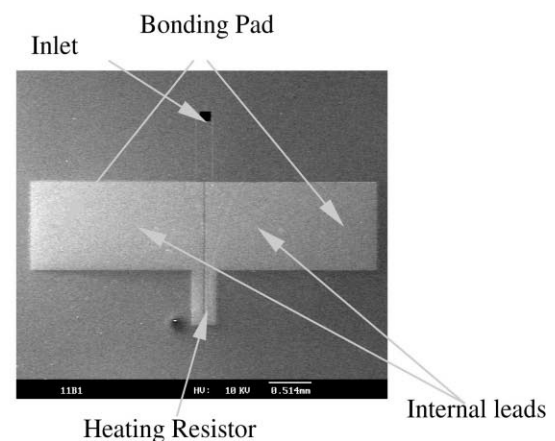
4. Tesing

The performances of the micro-thrusters have been tested. Four types of micro-thrusters used for testing are described in Table 1.

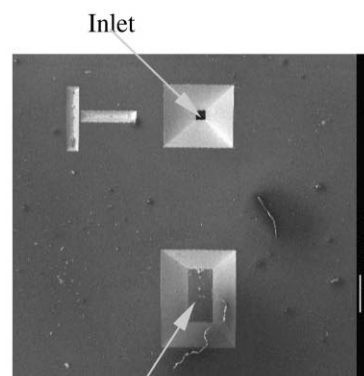
Using an electric balance whose precision is up to 0.1 mg, we recorded the mass decrement of the micro-thrusters and the water tank after operation for a period of time. In this way, we obtained the consumption rates of water propellant of the micro-thrusters. According to the test results, Series 3 has the smallest consumption rate. This means that the hydrophobic surface reduces propellant wasting. Fig. 8 shows the relationships of the consumption rate to voltage and to frequency, respectively for four micro-thrusters. From the Fig. 8, we can see that the consumption rate increases as



(a) The top wafer



(b) Front side of the bottom wafer



(c) Back side of the bottom wafer

Fig. 5. SEM graphs of a single micro-thruster chip: (a) the top wafer; (b) front side of the bottom wafer; (c) back side of the bottom wafer.

voltage and frequency are increased. The maximum consumption rate of propellant is 0.038 mg/s.

Fig. 9 shows the velocity and the diameter distributions of the ejected water vapor particles of a Series 1 micro-thruster, which are measured by a laser Doppler velocity

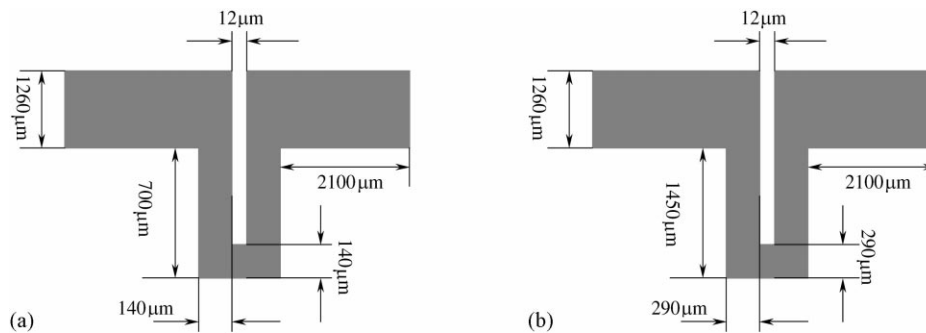


Fig. 6. Shapes of the metal Ti layers for two type of chambers.

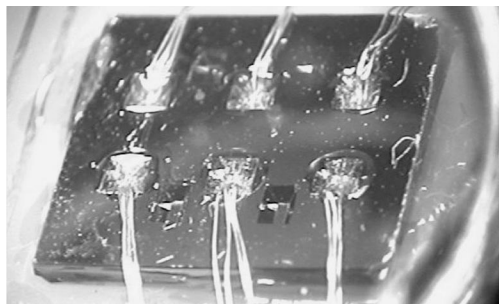


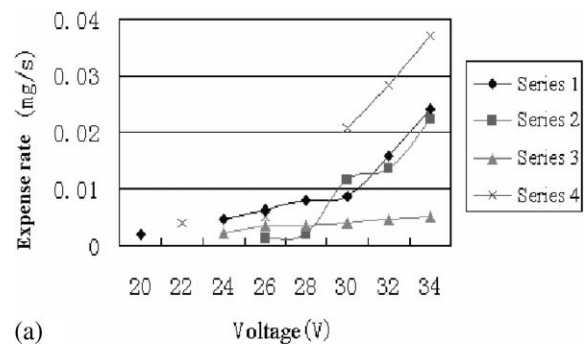
Fig. 7. The graph of a packaged 2 × 2 micro-thruster array.

measurement system. According to Fig. 9, the maximum velocity of the ejected water vapor particles is 20 m/s, the average velocity is 8.9095 m/s, the velocity variance is 4.81105 m/s, and the average diameter of the ejected water vapor particles is 19.5 μm, with a driving pulse width of 900 μs, a voltage of 34 V, and a frequency of 30 Hz. The total impulse of the micro-thruster produced in a second, calculated from the consumption rate and the average velocity of the ejected vapor, is more than 0.2×10^{-6} N s [5].

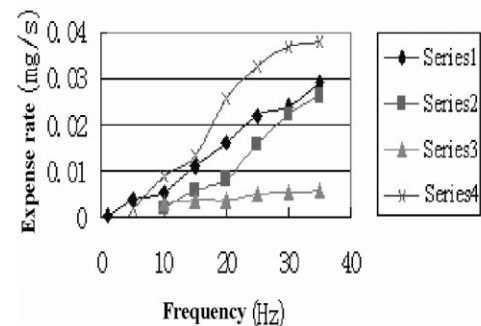
Table 2 shows the average velocities and the average diameters of ejected water vapor particles for the four types of the micro-thrusters versus the driving pulse width, and Table 3 shows these versus the driving voltage. The micro-thruster of Series 1 has a maximum velocity and that of Series 3 has a maximum diameter. The velocities of Series 2 and 4 are lower than those of the Series 1 and 2, because the chamber volume for Series 2 and 4 is four times larger than that for Series 1 and 3, but the driving power is only about two times larger. The average diameter of ejected water vapor particles decreases when the driving voltage changes

Table 1
Four types of micro-thrusters

Type of thrusters	Vaporizing chamber area (μm ²)	Nozzle shape	Heating resistor (Ω)
Series 1	300 × 750	Convergent	39
Series 2	600 × 1500	Convergent	36
Series 3	300 × 750	Convergent, hydrophobic	39
Series 4	600 × 1500	Divergent	36



(a)



(b)

Fig. 8. Relationships of flow rate to voltage (a) and to frequency (b).

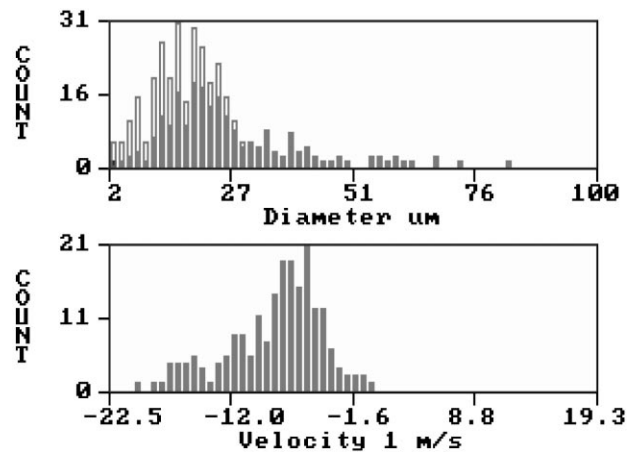


Fig. 9. Velocity and diameter distributions of ejected water vapor particles. Pulse width: 900 μs; voltage: 34 V; frequency: 30 Hz.

Table 2

Average velocities and diameters vs. driving pulse width for four types of the micro-thrusters

Pulse width (μs)	Series 1		Series 2		Series 3		Series 4	
	Velocity (m/s)	Diameter (μm)	Velocity (m/s)	Diameter (μm)	Velocity (m/s)	Diameter (μm)	Velocity (m/s)	Diameter (μm)
600	9.22	19.9						
700	9.57	17.6			6.68	25.7		
800	8.95	18.4			5.68	26.8		
900	8.91	19.5			4.47	31		
1000					5.04	26.5		
1100					4.23	15		
1200							3.77	19.1
1300			3.44	18.5			4.96	13.2
1400			3.55	15.2			4.61	22.4
1500			3.05	17.1			4.03	14.2

Table 3

Average velocities and diameters vs. driving voltages for four types of the micro-thrusters^a

Voltage (V)	Series 1		Series 2		Series 3		Series 4	
	Velocity (m/s)	Diameter (μm)	Velocity (m/s)	Diameter (μm)	Velocity (m/s)	Diameter (μm)	Velocity (m/s)	Diameter (μm)
32	6.43	21.2	3.57	17.2			2.1637	34.4
34	8.95	18.4	3.05	17.1	5.68	26.8	4.0318	14.2
36	8.58	14.8			5.65	19.7		
38					4.52	17.5		

^a Series 1, Series 3: frequency: 30 Hz, pulse width: 800 μs ; Series 2, Series 4: frequency: 30 Hz, pulse width: 1500 μs .

to high since the heating power is larger, the water is vaporized more completely. The maximum velocity of ejected water vapor particle is up to 28 m/s.

The thrust force of a micro-thruster is roughly determined using a measurement system shown in Fig. 10. The micro-thruster is mounted on the foundation of a micro translation stage, and a cantilever beam is fixed on the moving part of the stage which adjusts the free end of the cantilever beam to be aligned with the nozzle. When the micro-thruster is operating, the gas which dashes out through the nozzle hits the free end of the beam and makes it deflect.

By measuring the displacement of the free end of a cantilever beam, the approximate thrust of the micro-thruster can be calculated from the following formula:

$$F = \frac{3EI}{l^3}d \quad (1)$$

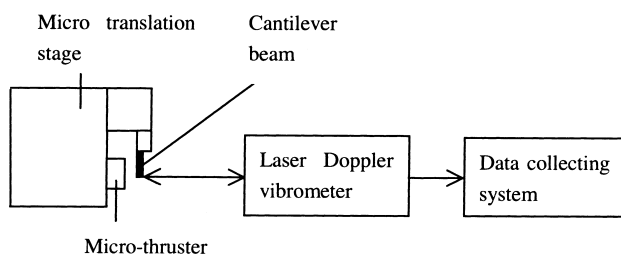


Fig. 10. Schematic diagram of a micro-thruster measurement system.

where E is the Young's modulus of the cantilever beam for metal Cu beam (120 GPa), l the length of the beam in the present case (5 mm), and I the cross-sectional moment of inertia of the beam, and can be expressed as

$$I = \frac{bh^3}{12} \quad (2)$$

where b is the width of the beam (2 mm) and h the thickness (30 μm).

Fig. 11 shows the displacement curve of the Series 1 micro-thruster measured with a laser Doppler vibrometer. The maximum thrust is about 2.9 μN according to this curve. Tables 4 and 5 show the thrust forces versus driving frequencies and voltages, respectively for Series 1 and 3 of micro-thrusters. The thrust force increases along with an increase of the driving pulse width and voltage.

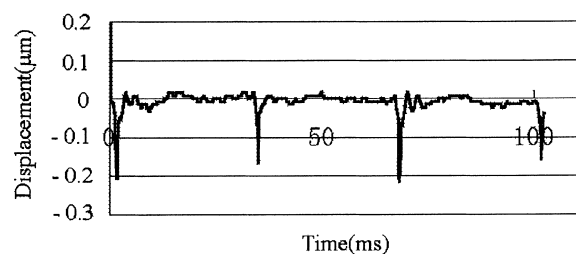
Fig. 11. Displacement of the free end of a cantilever. Pulse width: 900 μs ; voltage: 34 V; frequency: 30 Hz.

Table 4
Thrust forces vs. driving frequencies for two types of micro-thrusters^a

Pulse width (μs)	Thrust force (μN)	
	Series 1	Series 3
500	0.82	1.44
600	1.13	1.65
700	1.74	1.84
800	2.15	2.26
900	2.86	2.77

^a Driving frequency: 30 Hz, voltage: 34 V.

Table 5
Thrust forces vs. driving voltages for two types of micro-thrusters^a

Voltage (V)	Thrust force (μN)	
	Series 1	Series 3
32	0.71	1.02
34	2.15	2.26
36	2.57	2.57
38		2.77

^a Driving frequency: 30 Hz, pulse width: 800 μs .

5. Discussion

The cavity under the heating resistor reduced some heat loss. The thermal energy absorbed by the silicon substrate in one micro second, which is about the heating duration, is calculated using ANSYS. It is supposed that the heating resistor remains at unchange temperature of 300°C, the initial temperature of the silicon substrate is 20°C, and the environment around the wafer is heat-insulated. The silicon substrate with a size of 5000 $\mu\text{m} \times 5000 \mu\text{m} \times 520 \mu\text{m}$ has an etched cavity with a top size of 1300 $\mu\text{m} \times 1000 \mu\text{m}$ and a depth of 420 μm on the opposite side of the heating resistor at center of the substrate. The result of the evaluation shows that the cavity under the heating resistor can decrease the heat loss by 30%.

When the hot vapor escapes into the cold environment with normal temperature and atmospheric pressure, the vapor will coagulate, which reduces the velocity but increases the diameters of particles. As a result, the measured value of the velocity of ejected vapor is lower, compared with that in the real application.

The invacuum measurement condition, as well as the gap between the outlet of the nozzle and the free end of the beam, result in a measured value of the thrust less than the real one. Besides, although the actual action of the thrust is an impulse in a very short time, we calculated the thrust force as a static force, which also led to a lower value of the thrust.

Fig. 12 shows a micro-thruster nozzle after its operation for a few hours. It is visually found that the surface of the nozzle has been badly damaged, which will also reduce the performance and the lifetime of a thruster obviously. There-

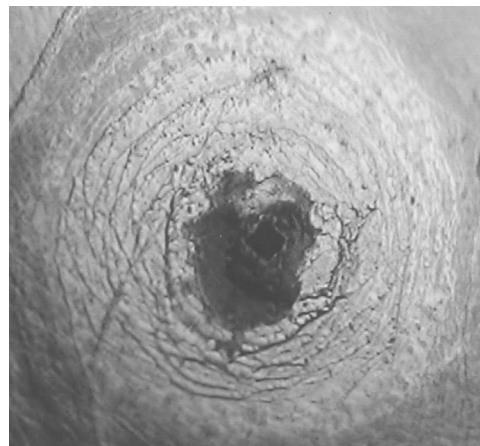


Fig. 12. Graph of a micro-thruster nozzle after the operation for a few hours.

fore, it will be necessary to coat a wearable layer such as a diamond-like film on silicon surface of nozzles to resist abrasion.

We temporarily selected water as the propellant because of its cheapness and safety, and its suitability for study in a lab. The result of the water propellant is significant for further investigation, though its impulsion and thrust are small. In practical applications, N_2H_4 , NH_3 , etc. can be selected as propellant for better performance.

6. Conclusion

In this paper, the design and configuration of prototypes of vaporizing water micro-thrusters have been described, and the primary test data have been reported. The application of the micro-thrusters aims at the position control of micro-satellites. The micro-thruster chip consists of a micro-resistor, a vaporizing chamber, a nozzle, an inlet and a micro channel, with two silicon wafers bonded together. In the test condition, the maximum average velocity of ejected vapor particles is 8.5 m/s, and the maximum velocity is 28 m/s. The maximum thrust is 2.9 μN and the total impulse produced in a second is more than $0.2 \times 10^{-6} \text{ N s}$. The maximum consumption rate of propellant is 0.038 mg/s. The study shows the micro-thrusters significant to the development of power MEMS and will be investigated more detail in future.

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